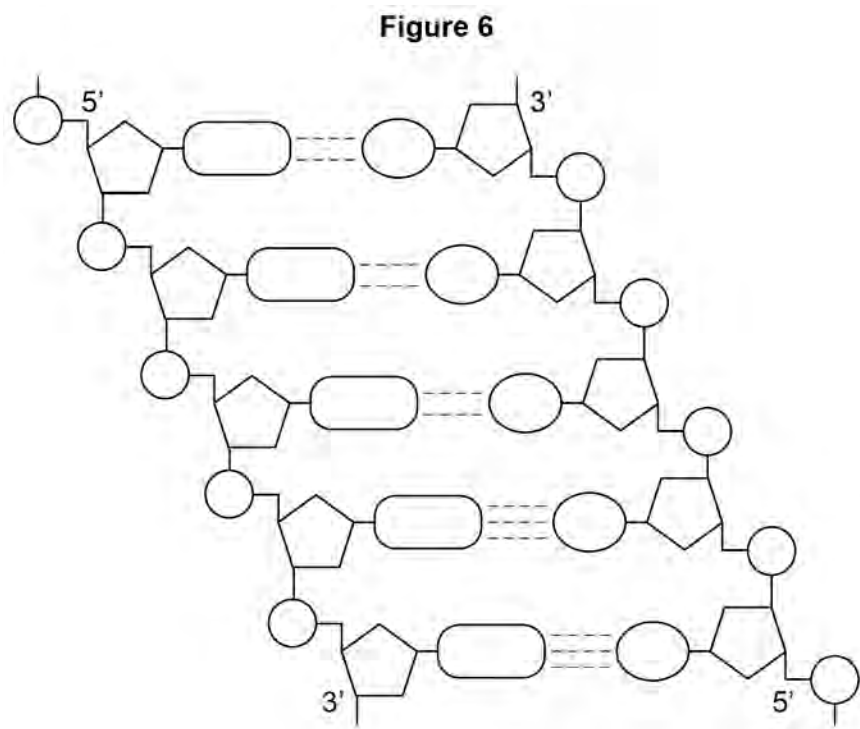


Figure 6 represents part of a DNA molecule.



0 8 . 1 Draw a box around a single nucleotide.

[1 mark]

Table 4 shows the percentage of bases in each of the strands of a DNA molecule.

Table 4

DNA strand	Percentage of each base			
	A	C	G	T
Strand 1	16			
Strand 2		21	34	

0 8 . 2 Complete Table 4 by adding the missing values.

[2 marks]

0	8	.	3
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During replication, the two DNA strands separate and each acts as a template for the production of a new strand. As new DNA strands are produced, nucleotides can only be added in the 5' to 3' direction.

Use **Figure 6** and your knowledge of enzyme action and DNA replication to explain why new nucleotides can only be added in a 5' to 3' direction.

[4 marks]

[Extra space]

Turn over for the next question

